

Equilíbrio Iônico (K_a , K_b , K_w , pH e pOH)

1- Ácidos



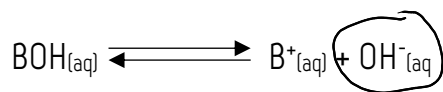
$$K_c = \frac{[\text{H}^+].[\text{X}^-]}{[\text{HX}]}$$

$\downarrow$
 K_a

↑ IONS ↑ molécula

$\uparrow K_a$ $\uparrow$ IONS $\uparrow \text{H}^+$ $\uparrow$ Acidez
 K_a elevado = mais ácido

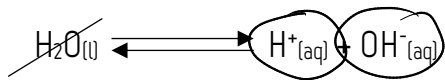
2- Bases



$$K_b = \frac{[\text{B}^+].[\text{OH}^-]}{[\text{BOH}]}$$

$\uparrow K_b$ $\uparrow$ IONS $\uparrow \text{OH}^-$ $\uparrow$ Básico
 K_b elevado = mais básico

3- Água



$$K_w = \frac{[\text{H}^+].[\text{OH}^-]}{[\text{H}_2\text{O}]}$$

$$K_w = [\text{H}^+].[\text{OH}^-] = 10^{-14}$$

Produto iônico da água

↑ condição amb.

4- pH x pOH

Definições

$$pK_a = -\log K_a$$

$$\downarrow \text{pH} = -\log[\text{H}^+] \uparrow$$

$$\log [\text{H}^+]^{-1}$$

ou

$$\log \frac{1}{[\text{H}^+]} \uparrow$$

$$\downarrow \text{pOH} = -\log[\text{OH}^-] \uparrow$$

$$[\text{H}^+] = 10^{-4} \text{ mol/L}$$

Exemplos

$$[\text{H}^+] = 10^{-4} \quad \text{pH} = -\log 10^{-4} = 4 //$$

$$[\text{OH}^-] = 10^{-6} \quad \text{pH} = 6$$

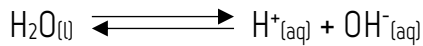
$$\text{pH} = 2 \quad [\text{H}^+] = 10^{-2} \text{ mol/L}$$

$$\text{pOH} = 10 \quad [\text{H}^+] = 10^{-10} \text{ mol/L}$$

Relação muito importante

$\text{pH} + \text{pOH} = 14$

Dedução



$$K_w = [\text{H}^+] \cdot [\text{OH}^-] = 10^{-14}$$

$$\log [\text{H}^+] \cdot [\text{OH}^-] = \log 10^{-14}$$

$$(-1) (\log [\text{H}^+], \log [\text{OH}^-]) = -14$$

$$-\log [\text{H}^+] \cdot -\log [\text{OH}^-] = -(-14)$$

$$\text{pH} + \text{pOH} = 14 //$$

5- Classificação das soluções

a) **Neutra** $[\text{H}^+] = [\text{OH}^-]$

$$[\text{H}^+], [\text{OH}^-] = 10^{-7}$$

$$10^{-7} \cdot 10^{-7} = 10^{-14}$$

$$\text{pH} = 7 \quad \text{e} \quad \text{pOH} = 7$$

b) **Ácida** $[\text{H}^+] > [\text{OH}^-]$

$$[\text{H}^+], [\text{OH}^-] = 10^{-14}$$

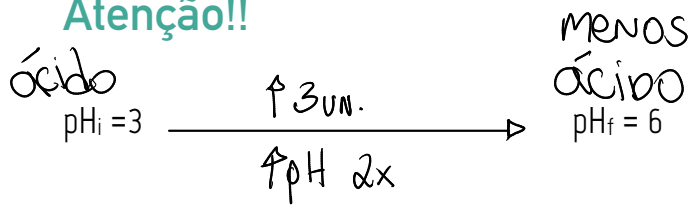
$$\text{pH} < 7 \quad \text{e} \quad \text{pOH} > 7$$

c) **Básica** $[\text{H}^+] < [\text{OH}^-]$

$$[\text{H}^+], [\text{OH}^-] = 10^{-14}$$

$$\text{pH} > 7 \quad \text{e} \quad \text{pOH} < 7$$

Atenção!!



$[\text{H}^+] = 10^{-3} \frac{\text{mol}}{\text{L}}$ $\xrightarrow{\downarrow 1000x}$ $[\text{H}^+] = 10^{-6} \frac{\text{mol}}{\text{L}}$

Anotações: